



Independent QCE practice

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# Practice Set A

2025 SYLLABUS EDITION

Question and response book

## Chemistry

### Paper 1

#### Time allowed

- Perusal time - 5 minutes
- Working time - 90 minutes

#### General instructions

- Answer all questions in this question and response book.
- A QCAA-approved calculator is permitted.
- The Chemistry formula and data book is provided.
- Planning paper will not be marked.

#### Section 1 (20 marks)

- 20 multiple choice questions

#### Section 2 (35 marks)

- 9 written response questions

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**THIS PAGE WILL NOT BE MARKED**

## Section 1

### Instructions

- This section has 20 questions and is worth 20 marks.
- Use a 2B pencil to fill in the A, B, C or D answer bubble completely.
- Choose the best answer for Questions 1-20.
- If you change your mind or make a mistake, use an eraser to remove your response and fill in the new answer bubble completely.

|                 | A                                | B                     | C                     | D                     |
|-----------------|----------------------------------|-----------------------|-----------------------|-----------------------|
| <b>Example:</b> | <input checked="" type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

|     | A                     | B                     | C                     | D                     |
|-----|-----------------------|-----------------------|-----------------------|-----------------------|
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| 20. | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

Ensure you have filled an answer bubble for each question.

Do not write outside this box.

## Section 2

### Instructions

- Write using black or blue pen.
  - Questions worth more than one mark require chemical reasoning and/or working to support the answer.
  - Give units, state symbols, significant figures and chemical notation where required.
  - If you need more space, use an additional page and identify the question number.
  - This section is worth 35 marks.
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**QUESTION 21** (3 marks)

Transition-metal oxidation states are represented by a signed number, while ionic compound names use a Roman numeral.

- a) Determine the oxidation state of cobalt in  $\text{Co}_2\text{O}_3$ . *[1 mark]*

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- b) Name  $\text{Co}_2\text{O}_3$ . *[1 mark]*

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- c) For  $2\text{Co}^{3+}(\text{aq}) + \text{Sn}^{2+}(\text{aq}) \rightarrow 2\text{Co}^{2+}(\text{aq}) + \text{Sn}^{4+}(\text{aq})$ , identify the oxidising agent. *[1 mark]*

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**QUESTION 22** (4 marks)

The table gives physical-property data for four organic compounds.

| Compound       | Class           | Molar mass<br>(g mol <sup>-1</sup> ) | Boiling point<br>(°C) | Water solubility |
|----------------|-----------------|--------------------------------------|-----------------------|------------------|
| but-1-ene      | alkene          | 56.1                                 | −6                    | very low         |
| pentane        | alkane          | 72.2                                 | 36                    | very low         |
| butan-1-ol     | alcohol         | 74.1                                 | 118                   | moderate         |
| propanoic acid | carboxylic acid | 74.1                                 | 141                   | very high        |

Physical properties

- a) Determine the most volatile compound. Justify your answer using the data. [2 marks]

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- b) Explain why butan-1-ol has a higher boiling point than pentane even though their molar masses are similar. [2 marks]

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**QUESTION 23** (4 marks)

A saturated calcium fluoride solution is described by  $\text{CaF}_2(\text{s}) \rightleftharpoons \text{Ca}^{2+}(\text{aq}) + 2\text{F}^{-}(\text{aq})$ . At  $25^{\circ}\text{C}$ , the saturated solution has  $[\text{F}^{-}] = 4.80 \times 10^{-4} \text{ M}$ .

- a) Calculate  $K_{sp}$  for calcium fluoride. [2 marks]

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- b) A small amount of calcium chloride is added without a significant change in volume. [2 marks]  
Predict whether a precipitate forms. Explain your reasoning.

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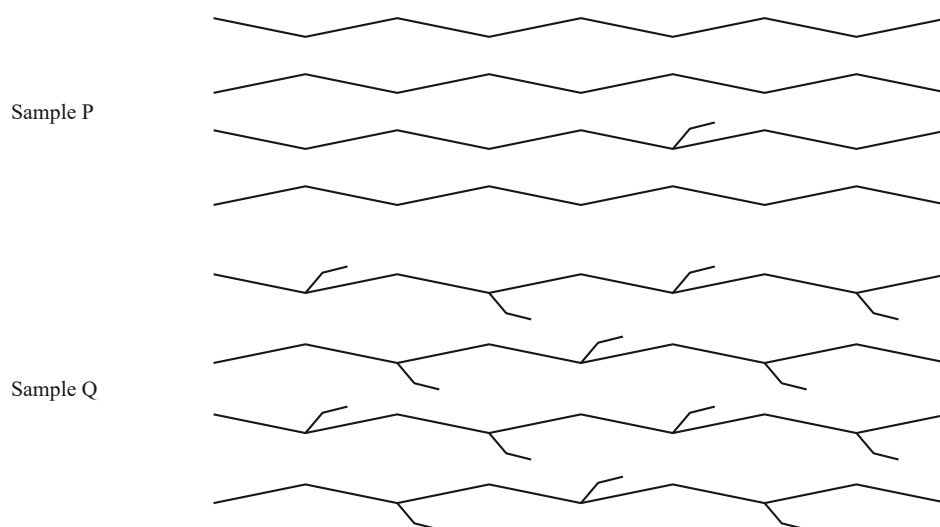
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**QUESTION 24** (4 marks)

Two polyethene samples have the chain arrangements shown.



Polyethene microstructures

- a) Identify the sample that represents HDPE and the sample that represents LDPE. [2 marks]

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- b) Explain why sample P has a greater density and generally greater strength than sample Q. [2 marks]

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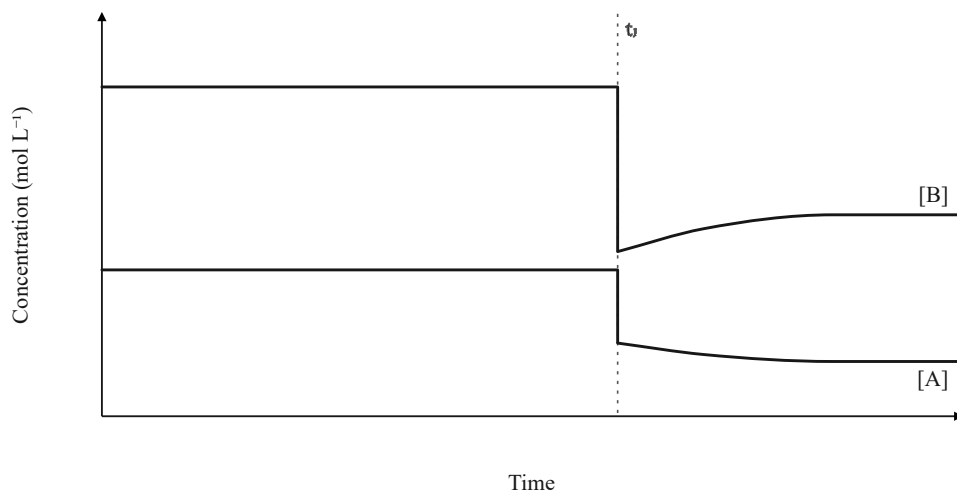
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### QUESTION 25 (4 marks)

The graph shows the concentrations in the equilibrium  $A(g) \rightleftharpoons 2B(g)$ . A single change is applied at  $t_1$  while temperature remains constant.



Concentration response to a change at  $t_1$

- a) Identify the change applied at  $t_1$ . [1 mark]

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- b) Explain the instantaneous changes in both concentrations at  $t_1$ . [1 mark]

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- c) Explain the subsequent changes as the system establishes a new equilibrium. [2 marks]

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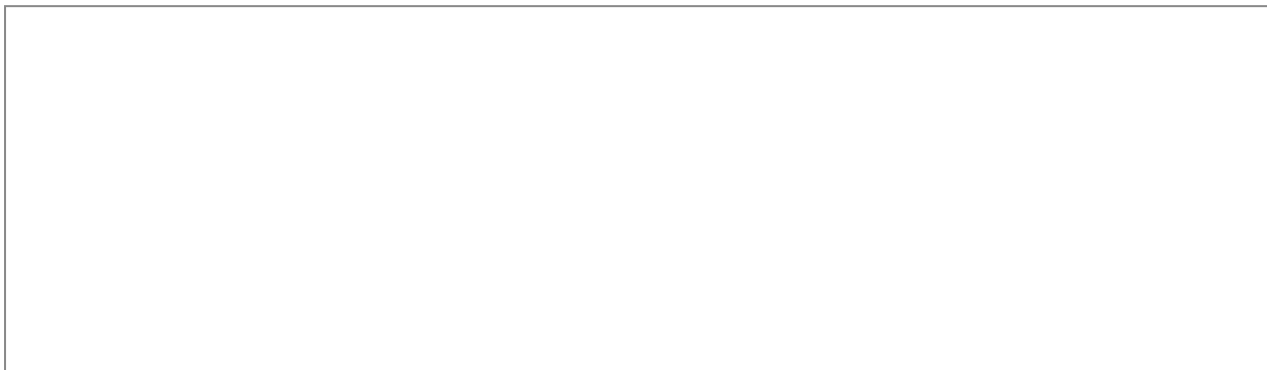
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**QUESTION 26** (4 marks)

But-2-ene has the molecular formula  $C_4H_8$ .

a) Draw both geometrical isomers of but-2-ene.

[2 marks]



b) Describe the result when either isomer is shaken with bromine water. Explain the observation.

[2 marks]

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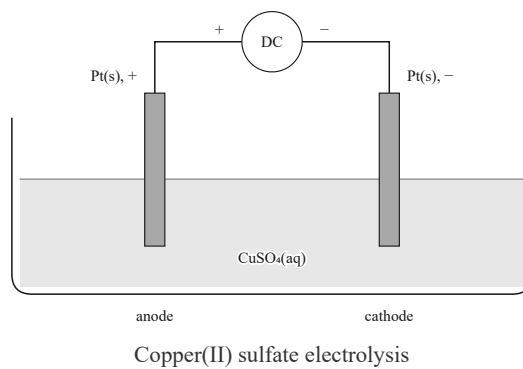
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**QUESTION 27** (4 marks)

Aqueous copper(II) sulfate is electrolysed using inert platinum electrodes. A current of 1.50 A passes for 1930 s.



- a) Write the cathode half-equation. Include states. [1 mark]

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- b) Calculate the mass of copper deposited. [3 marks]

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**QUESTION 28** (4 marks)

Ethanol is produced by hydration of ethene:  $\text{C}_2\text{H}_4(\text{g}) + \text{H}_2\text{O}(\text{g}) \rightarrow \text{C}_2\text{H}_5\text{OH}(\text{g})$ . A plant must collect 18.4 kg of ethanol from a process operating at 80.0% yield.

Calculate the minimum mass of ethene that must be supplied to the process. Assume water is in excess. *[4 marks]*

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**QUESTION 29** (4 marks)

Four 0.0100 M solutions are hydrochloric acid, ethanoic acid, sodium hydroxide and ammonia. The labels have been removed. Use all relevant evidence to identify the solutions.

| Solution | pH    | Conductivity |
|----------|-------|--------------|
| P        | 2.00  | high         |
| Q        | 3.38  | low          |
| R        | 12.00 | high         |
| S        | 10.62 | low          |

Measurements for four unknown solutions

Match P, Q, R and S to the four substances. Explain your assignments using pH, conductivity and degree of ionisation at equilibrium.

[4 marks]

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**END OF PAPER**

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### ADDITIONAL PAGE FOR STUDENT RESPONSES

Write the question number you are responding to.

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Paper 1 question and response book  
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